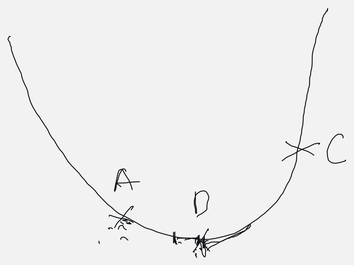


Slope at
A = -ve
B = 0
C = +ve

rate of change of slope = +ve

$\underline{\text{Slope}} = \frac{dy}{dx} = \frac{d^2y}{dx^2}$

$y = 10, \frac{dy}{dx} = 0$

$y = x \quad \frac{dy}{dx} = 1$

$y = 5x \quad \frac{dy}{dx} = 5$

$y = 5x + 10 \quad \frac{dy}{dx} = 5$

$y = x^2, \quad \frac{dy}{dx} = 2x$

$y = x^2 - 5x + 10 \quad \frac{dy}{dx} = 2x - 5$

$y = (x+10)^2 \quad \frac{dy}{dx} = 2(x+10)$

$y = (2x+5)^2 \quad \frac{dy}{dx} = 2(2x+5) * 2 = 8x + 20$

$y = 10m + 2c + 30$

$\frac{\partial y}{\partial m} = 10, \quad \frac{\partial y}{\partial c} = 2$

$y = (2m + c - 5)^2 + (3m + c - 10)^2$

$\frac{\partial y}{\partial m} = 2(2m + c - 5) \times 2 + 2(3m + c - 10) \times 3$

$= 8m + 4c - 20 + 18m + 6c - 60 = 26m + 10c - 80$

$\frac{\partial y}{\partial c} = 2(2m + c - 5) + 2(3m + c - 10)$

$= 4m + 2c - 10 + 6m + 2c - 20 = 10m + 4c - 30$

We need to find m, c for which y is minimum

$\frac{dy}{dm} = 0, \quad \frac{dy}{dc} = 0$

$26m + 10c - 80 = 0$
$10m + 4c - 30 = 0$

$m = 5$
$c = -5$

$y = (2x + 5 - 10)^2 + (3x + 4 - 20)^2$

$\frac{dy}{dx} =$

$2(2x + 5 - 10) \times 2 + 2(3x + 4 - 20) \times 3$

$26x - 116$

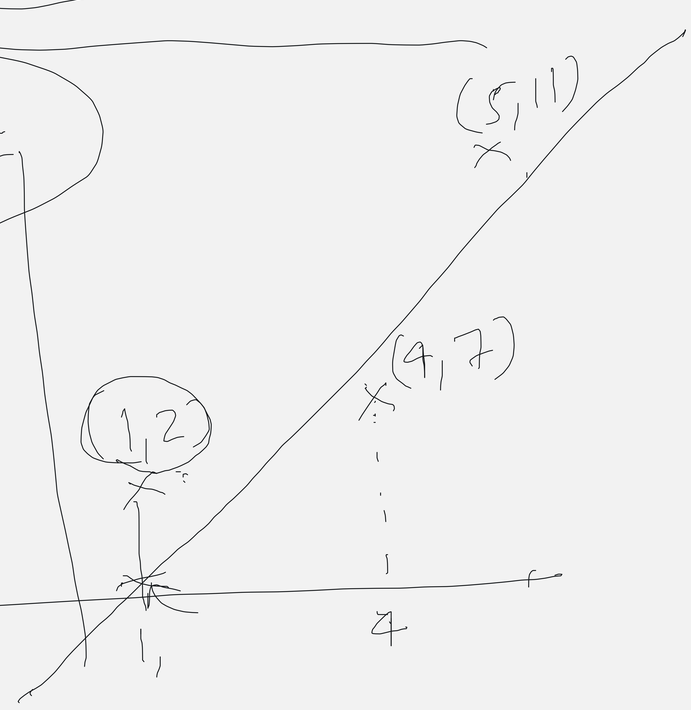
$26x - 116$

① Assume $y = mx + c$

② $E_1 = -m * 1 + c = 2$.

$E_2 = 4m + c - 7$.

$E_3 = 5m + c - 11$.



$E = E_1^2 + E_2^2 + E_3^2$

$E = (m + c - 2)^2 + (4m + c - 7)^2 + (5m + c - 11)^2$

Find m, c for which E is minimum